them, and continues from the top-ranked harnesses on the selection tasks through iterative beam search. The harness it discovers, AutoRef-Harness, makes the open-weight FLUX.2 [klein] 4B competitive with proprietary models on MultiBanana, transfers to unseen reference counts, an unseen benchmark and evaluator, other generators, and another reasoning model, and adds to the gains of fine-tuning. These results suggest that how frozen models are used is itself worth optimizing.

AI USE STATEMENT

In this work, we used generative AI tools for the following tasks: *design or provide feedback on research methodology or experiments, implement methods, assist with translation, and support qualitative and thematic data analysis*. We have not used generative AI tools for the following tasks: *help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, propose or refine hypotheses, clean and reformat datasets, interpret results*, and the tasks *generate synthetic data sets* and *assist in the writing of proofs* are not applicable to this work. Additionally, we used generative AI tools for the following tasks: *create or modify scientific figures or images, create or edit software code, draft parts of a research paper, and edit a research paper to improve readability*. We have reviewed all AI-assisted work. Two authors verified and tested LLM-generated code and writing for correctness. We take responsibility for the final content, including text, claims, or artifacts produced with generative AI.

ETHICS STATEMENT

Our method optimizes only the agent harness and does not update the underlying reasoning or image generation models. Thus, it does not introduce new behaviors through model-weight modification, although it can change how existing model capabilities are composed. As with image generation more broadly, improved multi-reference image generation may be misused to create misleading or unauthorized synthetic content. The harness also inherits limitations and biases from the underlying models and evaluators. We therefore encourage responsible use of the released code and harness.

REPRODUCIBILITY STATEMENT

We release the AutoRef implementation and AutoRef-Harness at an URL (https://github.com/KuOnoda/AutoRef), together with the data splits and the pinned model revisions. We conduct all evaluations on publicly available benchmarks. The main paper describes the harness optimization procedure, model configurations, data splits, and evaluation protocol, while the Appendix provides additional evaluation details and complete results.

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REFERENCES

Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alex Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. Gepa: Reflective prompt evolution can outperform reinforcement learning. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust (eds.), *International Conference on Learning Representations*, volume 2026, pp. 8479–8565, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/0e9e708b6f48e14fd0ac29e167413f76-Paper-Conference.pdf.

Anthropic. Claude code. https://claude.com/product/claude-code, 2025. Accessed: 2026-09-18.

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, Mei Li, Kaixin Li, Zicheng Lin, Junyang Lin, Xuejing Liu, Jiawei Liu, Chenglong Liu, Yang Liu, Dayiheng Liu, Shixuan Liu,

M. Ghassemi, and N. Chen (eds.), *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pp. 45128–45164. Curran Associates, Inc., 2025. doi: 10.52202/085713-1503. URL https://proceedings.neurips.cc/paper_files/paper/2025/file/40168e00bf87869c5d153e934d8a3602-Paper-Conference.pdf.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun (eds.), *International Conference on Learning Representations*, volume 2024, pp. 34133–34156, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/9156b0f6dfa9bbd18c79cc459ef5d61c-Paper-Conference.pdf.

Yaru Hao, Zewen Chi, Li Dong, and Furu Wei. Optimizing prompts for text-to-image generation. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 66923–66939. Curran Associates, Inc., 2023. doi: 10.52202/075280-2923. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/d346d91999074dd8d6073d4c3b13733b-Paper-Conference.pdf.

Haoran He, Jiajun Liang, Xintao Wang, Pengfei Wan, Kun Gai, and Ling Pan. Scaling image and video generation via test-time evolutionary search, 2026a. URL https://openreview.net/forum?id=CFlOUNWsaP.

Zefeng He, Siyuan Huang, Xiaoye Qu, Yafu Li, Tong Zhu, Yu Cheng, and Yang Yang. GEMS: Agent-native multimodal generation with memory and skills. *arXiv preprint arXiv:2603.28088*, 2026b.

Junyao Hu, Zhongwei Cheng, Waikeung Wong, and Xingxing Zou. Garments2look: A multi-reference dataset for high-fidelity outfit-level virtual try-on with clothing and accessories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Shengran Hu, Cong Lu, and Jeff Clune. Automated design of agentic systems. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (eds.), *International Conference on Learning Representations*, volume 2025, pp. 21344–21377, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/36b7acf6f6010652b3f2a433774a66fe-Paper-Conference.pdf.

Chengsong Huang, Zifeng Wang, Rujun Han, Jun Yan, Yanfei Chen, Zoey CuiZhu, Ke Jiang, Peng Xia, Han Yu, Yufan Zhuang, Yifei Ming, Jiaqi Pan, Bhavana Dalvi Mishra, Jiaxin Huang, Burak Gokturk, Tomas Pfister, and Chen-Yu Lee. Envharness: Awakening static worlds for agent learning, 2026a. URL https://arxiv.org/abs/2608.19880.

Wenwang Huang, Yusen Fu, Junjie Wang, Mengfei Huang, Yulin Li, Gan Liu, Jing Cai, Yancheng He, and Zhuotao Tian. Scaling multi-reference image generation with dynamic reward optimization. In *The 19th European Conference on Computer Vision*, 2026b.

Naoto Inoue, Kotaro Kikuchi, Edgar Simo-Serra, Mayu Otani, and Kota Yamaguchi. LayoutDM: Discrete Diffusion Model for Controllable Layout Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10167–10176, 2023.

Kaixun Jiang, Yuzheng Wang, Junjie Zhou, Pandeng Li, Zhihang Liu, Chen-Wei Xie, Zhaoyu Chen, Yun Zheng, and Wenqiang Zhang. Genagent: Scaling text-to-image generation via agentic multimodal reasoning. *The 19th European Conference on Computer Vision*, 2026.

Jaemin Jung, Kyeongha Rho, Inkyu Shin, and Joon Son Chung. Inference-time scaling for joint audio-video generation. *Transactions on Machine Learning Research*, 2026. ISSN 2835-8856. URL https://openreview.net/forum?id=MHNFjjm5nO.

Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan A, Saiful Haq, Ashutosh Sharma, Thomas Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, and Christopher Potts. Dspy: Compiling declarative language model calls into state-of-the-art pipelines. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun